

token IDs / sequence

hidden vector sequence

next token probabilities

Token Processing

$$t_i \rightarrow h_i$$

2. Token embedding



3. RoPE position encoding

1. Input token sequence

LLaMA3-style Backbone

causal Transformer, N layers

4. N stacked Transformer blocks

RMSNorm

GQA

SwiGLU FFN



5. RMSNorm (Final)

Language Modeling

$$p(t_{next} | h)$$

6. Linear LM head: ar_lm_head



7. Next-token logits

8. Sampled next token